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Singly Linked List:

A single linked is a type of linear data structure where data is stored in separate nodes.
Each node have two parts.

Stored value in node. ← [data | Next Pointer] → a address of next node

Elements of Singly Linked List.

1 **Node:**
A node is a single element of a linked list. Each node has two parts. First part for value or data and next part for address.

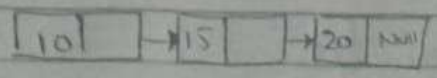
[Pointer | Next]

2. **Data Field:**
A part of the node that stores the actual value or data associated with that element in the list.
Example:

[20]

3. **Next pointer:**
- Stores address of next node.
- Connects one node to another.
- last node points to null.

Example:



* 10 points to 15
* 15 points to 20
* 20 points to Null

4 Head Pointer:

- Stores address of first node.
- Starting point of linked list.

Example:

Part A- INSERTION OPRATION

1 Insertion at beginning:

insert 10 into 20→30→40

- Initial list.

  Head→ 20→30→40 →Null.

- Pseudocode:

```
CREATE newNode
newNode.data = 10
NewNode.next = head
head = newNode
```

- Dry Run:

Step 1: Create newNode (10)

Step 2: new Node.next = head
Therefore 10 points to 20
Step 3: head = New Node.

- Pointer Movement:
  Before: Head -> 20
  After: Head -> 10

- Final Linked List:
  Head -> 10 -> 20 -> 30 -> 40 -> Null

2 Insertion at End:

Insert 50 into : 10 -> 20 -> 30

- Initial linked list:
  Head -> 10 -> 20 -> 30 -> Null

- Pseudocode

```
CREATE newNode
 newNode.data = 50
 newNode.next = Null

 temp = head
 WHILE temp.next != Null
        temp = tem.next
 temp.next = newNode
```

- Dry Run:

Step 1: create newNode (50)
Step 2: Initialize temp

temp → 10 → 20 → 30 → Null.

Step 3: First Iteration of while.

10.next = 20 → Not Null

Step 4: Secound Iteration

20.next = 30 → Not Null

temp = temp.next.

Step 5: Third check

30.next = Null

loopstop here.

Step 6: Attach new node.

30 → 50

- Pointer movement: Before 20→30 After 30→50
- Final linked

Head → 10 → 20 → 30 → 50 → Null

3. Insertion at Specific Position:

Insert 25 at position 3 into list:

10 → 20 → 30 → 40.

- Initial list:

Head → 10 → 20 → 30 → 40 → Null.

- Pseudocode.

```
CREATE newNode
newNode.data = 25
temp = head
```

For i=1 to position-2
temp = temp.next

NewNode.next = temp.next
temp.next = nextNode

- Dry Run

Step 1: creat new Node
newNode = Node
NewNode.data = 25
Memory allocated For newnode.
newNode → [25 | Null]
Step 2: Start From head
temp = head
temp points to the First node.
temp → [10 |]
Step 3: Move to the node before the insertion point
position = 3 so position-2=1.
loop runs once.
Step 4: link newNode to the next Node
newNode → 25 points to 30.

Step 5: link previous node to newNode.
10 → 20 → 25 → 30 - Null

- Pointer movement:
Before 20 → 30

After: 20 → 25

- Final linked
  Head → 10 → 20 → 25 → 30 → 40 → Null.

PART. B - DELETION OPERATIONS

1. Deletion From Beginning

Delete First node Frist node From list.

10 → 20 → 30

- Initial list:
  Head → 10 → 20 → 30 → Null

- Pseudocode

```
temp = head
head = head.next
Delete temp.
```

- Dry run:
  Step 1: temp = 10
  Step 2: Move head
  Head → 20
  Step 3: Delete 10.

- Pointer movement:
  Before:
  10 → 20
  After:
  20 → 30

- Final linked:

  Head → 20 → 30 → Null

2. Deletion From the End:

Delete last node from list:

10 → 20 → 30 → 40.

- Initial linked list:

  Head → 10 → 20 → 30 → 40 → Null

- Pseudocode

```
temp = head
WHILE temp.next != Null
  temp = temp.next.
Delete temp.next.
temp.next = Null
```

- Dry Run:

Step1: tem = 10

Step2: tem → 20

Step3: tem → 30 (stope here)

Step4: Delete 40

- Pointer movement:

  Before:

  40 → Null

  After:

  30 → Null

- Final linked list:

  Head → 10 → 20 → 30 → Null

3 Deletion at Specific Position:

Delete position 2 from list:

10 → 20 → 30 → 40

- Pseudocode.

```
temp = head
For i = 1 to position - 2
temp = temp.next
nodeToDelete = temp.next
temp.next = nodeToDelete.next
DELETE nodeToDelete
```

- Dry Run

Step 1: temp now points to 10

Step 2: Move to one node before the one to delete.

Step 3: Identify the node to delete.

- temp.next point to 30
- nodeToDelete now points to 30

Step 4: Update links to skip the delete node.

- nodeToDelete.next is 40
- So, now points to 40.

Step 5: Delete 30.

Pointer movement:

Before:

20 → 30 → 40 → Null

After:

20 → 40 → Null

- Final linked list:

Head → 10 → 20 → 40 → Null